

Delay-Aware UAV Computation Offloading and Communication Assistance for Post-Disaster Rescue

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Abstract—In this paper, we consider an unmanned aerial vehicle (UAV)-assisted post-disaster rescue scenario, where UAV-mounted aerial base stations (ABSs) compute tasks related to post-disaster rescue operations while also providing communication services to ground users (GUs). With the limited computation capacity of ABSs, we aim to minimize the task computation queuing delay and ensure the GU communication rate by jointly optimizing ABS-GU association, task offloading, and ABS trajectory. The problem is formulated as a mixed-integer nonlinear program, and a solution is proposed by integrating Lyapunov optimization and actor-critic based deep reinforcement learning. We utilize a model-based successive convex approximation technique in a critic module to acquire an accurate evaluation of actor module output. Simulation results demonstrate the effectiveness of the proposed approach in reducing the task computation queuing delay.

Index Terms—Unmanned aerial vehicle (UAV), computation offloading, resource allocation, Lyapunov optimization.

I. INTRODUCTION

TERRESTRIAL infrastructure damages bring severe communication outage when extreme weather or emergencies happen [1]. The communication between macro-cell base station (MBS) and local ground users (GUs) can be blocked in a disaster environment due to the sudden interruption of wireless services. In post-disaster rescue, emergency communications are required to support GUs sending disaster information for

timely and effective rescue operations. Meanwhile, computing capabilities are essential for using sensors to detect GUs' location information through bio-signals or audio signals. Due to high mobility and flexible deployment, unmanned aerial vehicles (UAVs) embedded with on-board sensors can potentially enter the hard-to-reach and affected areas [2]. As such, UAVs are not only deployed to establish emergency communication infrastructure in the air, referred to as aerial base stations (ABSs), but also can provide situational awareness via sensing the post-disaster rescue environment. However, in current studies on ABS-assisted post-disaster rescue, it is common to provide either emergency communication or situational awareness services separately by an ABS, rather than simultaneously provisioning both services. Typically, communication targets and sensing targets are different, and the ABS must approach different targets to provide high-speed communication services or high-precision sensing services, resulting in different flight trajectories. As a result, the existing work will not be able to provide both services simultaneously. Nonetheless, combining two services can improve the timeliness and resource utilization of post-disaster rescue services. However, due to the limited onboard battery and computational capability of ABSs, real-time computation tasks based on collected sensing data, such as image processing, cannot be fully supported by the ABSs with performance satisfaction [2], [3], [4], [5], [6], [7], [8]. Consequently, the time required to execute search and rescue operations may be prolonged and the efficiency of disaster rescue may be degraded [9]. The MBS has stronger computing capabilities compared to ABS and can provide computing support for ABS. Therefore, the computation task of ABS can be partially offloaded to the MBS for timely rescue responses. The task computed at ABS or MBS has to wait for the service completion of its preceding tasks in the queue, resulting in a task queuing delay. In this paper, we consider an ABS-assisted post-disaster rescue system, in which the ABS flies according to a designed trajectory, while providing communication services for associated GUs and collecting sensing data of the rescue environment. Meanwhile, the ABS partially offloads its computation tasks to the MBS for queuing delay improvement.

There are several challenges in ABS-GU association, task offloading, and trajectory planning. First, GUs are not uniformly distributed and the number of GUs in the coverage area of an ABS varies quickly due to the mobility of

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ABS [10], [11]. Hence, it is necessary to design an ABS-GU association strategy with the consideration of user distribution, ABS mobility, and ABS trajectory. Second, the channel state between ABS and MBS is time-varying due to the mobility of ABS [12]. It is difficult for the ABS to make optimal task offloading decisions based on time-varying channel state, which leads to undesired long task queuing delays. Third, the ABS simultaneously provides communication service to GUs and partially offloads computation tasks to MBS while flying. The onboard energy of an ABS is usually limited and a significant portion of ABS energy consumption stems from mechanical actions during flying. Providing communication services requires the ABS to get close to GUs to improve the throughput, while offloading tasks require the ABS to get close to the MBS to reduce the task offloading delay. Due to the inconsistent requirements between providing communication service to GUs and offloading computation tasks to the MBSs, the ABS trajectory design should strike a balance that accommodates both providing communication service and offloading computation tasks. Moreover, the ABS trajectory influences the channel states from the ABS to both GUs and MBSs, which are important factors in making the ABS-GU association and task offloading decisions. It can be seen that ABS-GU association, task offloading, and ABS trajectory are coupled in this system, which makes the system management even more complex. Therefore, we need to jointly optimize the ABS-GU association, ABS trajectory, and task offloading ratio under the ABS energy constraints to minimize task computation queuing delay and ensure the communication requirements of GUs.

Jointly optimizing the ABS-GU association, task offloading, and ABS trajectory planning is computationally complex, especially in a multi-GU and multi-ABS network [13]. Some existing studies based on optimization algorithms focus on designing sub-optimal solutions with reduced complexity, but this can lead to performance losses. Moreover, when the network environment changes, these algorithms recalculate the joint decisions based on the current conditions, which is time-consuming [14], [15], [16], [17]. The recent development of data-driven deep reinforcement learning (DRL) provides a promising alternative to reduce the computational complexity [18], [19]. Once the training of a DRL model is finished, real-time decisions can be made based on fast model inference as the environment changes [20], [21], [22], [23]. However, it is difficult to rely on only DRL-based methods for addressing the long-term performance requirements in a random environment.

In this work, we formulate the joint ABS-GU association, task offloading, and ABS trajectory planning problem as a long-term stochastic mixed integer nonlinear programming (MINLP) problem. To solve this problem, we propose a joint DRL and Lyapunov optimization (JDL) algorithm. We apply Lyapunov optimization to decouple the long-term stochastic MINLP into a series of deterministic MINLP problems. For each deterministic MINLP problem, we propose a solution that integrates model-based optimization and model-free DRL based on the actor-critic structure. The actor module is a deep neural network (DNN) that learns the optimal ABS-GU

association action based on the channel gains of all GUs and task queuing delays of all ABSs. The critic module evaluates the ABS-GU association action by analytically solving a joint ABS trajectory planning and task offloading problem based on the successive convex approximation (SCA) technique. The joint ABS trajectory planning and task offloading problem is decoupled to an ABS trajectory planning subproblem and a task offloading subproblem. To reduce the computation complexity, the trajectory planning subproblem is optimized on a large timescale, and the task offloading subproblem is optimized on a small timescale. The main contributions of this work are summarized as follows.

- 1) We develop a model for a joint ABS-GU association, ABS trajectory planning, and task offloading optimization problem in the ABS-assisted post-disaster rescue system, which is formulated as a stochastic MINLP. The model incorporates GU communication service guarantee under ABS energy constraints with the objective of minimizing queuing delay of sensing-related computation tasks.
- 2) We exploit Lyapunov optimization to decouple the long-term stochastic MINLP into a series of deterministic MINLP problems. For each deterministic MINLP problem, we propose a solution that integrates model-based optimization and model-free DRL based on the actor-critic structure. Compared to a conventional actor-critic structure that uses a model-free DNN in the critic module, we apply the SCA technique to acquire an accurate evaluation of an ABS-GU association action, which is the output of actor module.

The remainder of this paper is organized as follows. Related works are discussed in Section II. The system model is presented in Section III. Problem formulation and the solution are given in Sections IV and V, respectively. Finally, extensive simulations are discussed in Section VI, followed by conclusions in Section VII. For easy reference, Table I summarizes main mathematical symbols.

II. RELATED WORK

A. ABS-Assisted Communication for Post-Disaster Rescue

The ABS communication has been studied to assist post-disaster rescue. Zhao *et al.* establish a unified framework for an ABS-assisted emergency network in disasters, which jointly optimizes the trajectory and scheduling of ABSs to provide wireless service to ground devices [1]. In [24], a generic framework for disaster resilient communication is investigated, utilizing terrestrial edge nodes and ABSs to deal with the infrastructure disruption and surging workload. In [25], Dai *et al.* investigate the channel allocation and data delivery problems to provide communication and data delivery services for affected users. In [26], Wang *et al.* propose a joint MBS power allocation, ABS service zone selection, and user scheduling solution based on DRL to forward information from MBS to GUs through ABS, while enhancing the total spectrum efficiency. In [27], Feng *et al.* propose a joint ABS deployment and resource allocation scheme based on deep Q-network (DQN) for a multi-ABS enabled non-orthogonal multiple access system to extend the ABS coverage for GUs.

TABLE I
SUMMARY OF SYMBOLS

Symbol	Definition
A_j, M_g, U_i	j -th ABS, g -th MBS, i -th GU
δ_t	Time interval length
T	Time slot length
N	Entire task period
$l_g(t)$	Location of g -th MBS at time slot t
$l_j(t)$	Location of j -th ABS at time slot t
$l_i(t)$	Location of i -th GU at time slot t
l_j^{init}	Initial location of j -th ABS
$D_{\min}$	Minimum safe distance for any two ABSs
$v_{\max}$	Maximum ABS flight velocity
$v_{\min}$	Minimum ABS flight velocity
$\mathbf{v}_j(t)$	Velocity of j -th ABS at time slot t
$\mathbf{a}_j(t)$	Acceleration of j -th ABS at time slot t
$a_{\max}$	Maximum ABS flight acceleration
$\gamma_j(t)$	Task offloading ratio of ABS A_j at time slot t
$x_{ij}(t)$	Association strategy between ABS A_j and GU U_i at time slot t
N_A	Maximum GU association number of ABS in each time slot
$h_{ij}(t)$	Channel gain from ABS A_j to the GU U_i at time slot t
$h_{jg}(t)$	Channel gain from ABS A_j to the MBS M_g at time slot t
$r_{ij}(t)$	Transmission rate of between the ABS A_j and GU U_i at time slot t
$r_{jg}(t)$	Transmission rate of between the ABS A_j and MBS M_g at time slot t
$Y_j(t)$	Computation task of ABS A_j at time slot t
$Y_j^L(t)$	Computation task $Y_j(t)$ executed locally at ABS at time slot t
$Y_j^O(t)$	Computation task $Y_j(t)$ offloaded to the MBS at time slot t
$C_j(t)$	Data size of computation task $Y_j(t)$ at time slot t
$C_j^L(t)$	Data size of computation task $Y_j^L(t)$ at time slot t
$C_j^O(t)$	Data size of computation task $Y_j^O(t)$ at time slot t
$Q_j^L(t)$	Local sub-tasks computation queue of ABS A_j at time slot t
$Q_j^O(t)$	Offloaded sub-tasks transmission queue of ABS A_j at time slot t
$Q_j^S(t)$	Offloaded sub-tasks computation queue of ABS A_j at time slot t
$D_j^L(t)$	Local computation bits of ABS A_j
$D_j^S(t)$	Computation bits at MBS M_g for ABS A_j
$f_j(t)$	CPU-cycle frequency of ABS A_j
L_j	Required CPU cycles per bit for computation
$p_j(t)$	Transmit power of ABS A_j at time slot t
$f_{jg}(t)$	MBS M_g Allocated CPU-cycle frequency to ABS A_j at time slot t
$T_j^L(t)$	Local task computation queuing delay of ABS A_j
$T_j^O(t)$	Offloaded task transmission queuing delay of ABS A_j
$T_j^S(t)$	Offloaded task computation queuing delay of ABS A_j
$E_j^{\text{fl}}(t)$	Flight energy consumption of ABS A_j at time slot t
$E_{\max}$	Maximum flight energy consumption of the ABS

The existing works consider resource allocation and trajectory planning in an ABS-assisted communication network supported by hovering ABSs. To provide communication service for spatially distributed GUs, our work studies ABS-GU association and trajectory planning of constantly moving ABSs, while considering dynamic channel conditions and mechanical operation constraints. An uneven and unknown GU distribution is considered in this study.

B. ABS Computation Offloading

The ABS computation offloading has been extensively investigated [3], [5], [13], [28], [29], [30], [31]. In [13], Bi et al. propose an online computation offloading algorithm to maximize the network data processing capability subject to the long-term data queue stability and average power constraints. In [5], Xu et al. focus on the computing delay issue in mobile edge computing (MEC) systems assisted by multiple ABSs

with the goal of task completion time minimization. In [3], Zhang *et al.* consider the partial computation offloading, user association, and central processing unit (CPU) cycle frequency allocation for a multi-ABS assisted MEC system, in which the objective is to maximize computation efficiency for one GU. In [32], Messous et al. investigate the problem of offloading heavy computation tasks of ABSs to MBSs while achieving the best possible tradeoff between energy consumption, time delay, and computation cost. ABS computation offloading has been studied in post-disaster rescue scenarios, where ABSs generate rescue-related computation tasks and offload to MBSs [2], [9]. In [2] and [9], Wang et al. propose a tripartite dynamic cooperation framework among ABSs, unmanned ground vehicles (UGVs), and base stations, where multiple ABSs offload computation task to UGVs. In [18], Luo et al. propose a separated Q-network (SDQN) algorithm to jointly make optimal computation offloading decisions and flying orientation choices for multi-UAV cooperative target

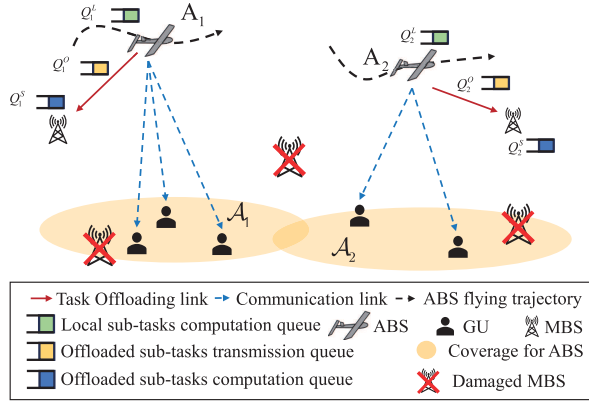


Fig. 1. An illustration of ABS-assisted communication network for post-disaster rescue.

search in an emergency network. Jointly considering ABS-assisted communication and ABS computation offloading is necessary for the ABS-assisted post-disaster rescue system. We investigate a joint ABS-GU association, ABS trajectory planning, and task offloading optimization problem with the objective to minimize the task computation queuing delay and ensure the GU communication rate.

III. SYSTEM MODEL

A. Network Model

We consider an operation cycle for ABS-assisted post-disaster rescue in a time duration of N seconds, which contains T discrete time slots with equal length $\delta_t = \frac{N}{T}$. Let $t = 1, \dots, T$ denote time slot index. The system is shown in Fig. 1, in which J ABSs have rescue-related computation tasks and provide communication services for I GUs. A number of G MBSs provide computation support for ABSs. Let $A_j (j \in \{1, \dots, J\})$ denote j -th ABS, $U_i (i \in \{1, \dots, I\})$ denote i -th GU, and $M_g (g \in \{1, \dots, G\})$ denote g -th MBS. Let $\mathcal{A}_j(t)$ denote ABS A_j 's coverage area at time slot t , where the coverage areas of different ABSs may overlap. GUs in area $\mathcal{A}_j(t)$ can connect to ABS A_j at time slot t . Let $s_{jg}(t)$ denote the association state between ABS A_j and MBS M_g at time slot t . ABS A_j is associated with MBS M_g when the distance between ABS A_j and MBS M_g is shorter than its distance to any other MBSs. We have $s_{jg}(t) = 1$ if ABS A_j is associated with MBS M_g at time slot t , and $s_{jg}(t) = 0$ otherwise.

B. ABS Trajectory Model

We consider a three-dimensional Cartesian coordinate system with MBS M_g , ABS A_j , and GU U_i at time slot t located at $\mathbf{l}_g(t)$, $\mathbf{l}_j(t)$, and $\mathbf{l}_i(t)$, respectively. In particular, the trajectory of ABS A_j is subject to velocity $\mathbf{v}_j(t)$ and acceleration $\mathbf{a}_j(t)$ constraints. Minimum safe distance $D_{\min}$ between ABSs must be kept to avoid collision. The trajectory coordinate of ABS A_j is subject to the following constraints:

$$\mathbf{l}_j(t+1) = \mathbf{l}_j(t) + \mathbf{v}_j(t)\delta_t + \frac{1}{2}\mathbf{a}_j(t)\delta_t^2, \quad \forall j, t \quad (1a)$$

$$\mathbf{v}_j(t+1) = \mathbf{v}_j(t) + \mathbf{a}_j(t)\delta_t, \quad \forall j, t \quad (1b)$$

$$\|\mathbf{l}_j(t) - \mathbf{l}_w(t)\|_2^2 \geq D_{\min}^2, \quad \forall j, w \in \{1, \dots, J\}, j \neq w, \quad \forall t \quad (1c)$$

$$\|\mathbf{v}_j(t)\| \leq v_{\max}, \quad \forall j, t \quad (1d)$$

$$\|\mathbf{v}_j(t)\| \geq v_{\min}, \quad \forall j, t \quad (1e)$$

$$\|\mathbf{a}_j(t)\| \leq a_{\max}, \quad \forall j, t \quad (1f)$$

where $v_{\max}$ and $v_{\min}$ denote the maximum and minimum ABS flight velocity, respectively, $a_{\max}$ denotes the maximum ABS flight acceleration, and $\|\cdot\|$ denotes the Euclidean norm. Constraints (1a)-(1f) represent ABS's trajectory constraints, including the maximum velocity constraint, minimum velocity constraint, maximum acceleration constraint, and minimum safe distance between any two ABSs.

C. ABS-GU Association Model

The ABS-GU association state between ABS A_j and GU U_i at time slot t is denoted as $x_{ij}(t)$. We have $x_{ij}(t) = 1$ if ABS A_j serves GU U_i at time slot t , and $x_{ij}(t) = 0$ otherwise. The ABS-GU association strategy is subject to the following constraints:

$$\sum_{j=1}^J x_{ij}(t) = 1, \quad \forall i, t \quad (2)$$

$$\sum_{i=1}^I x_{ij}(t) \leq N_A, \quad \forall j, t. \quad (3)$$

Constraint (2) indicates that each GU can be served by one ABS and constraint (3) indicates that each ABS can be associated with at most N_A GUs at each time slot t . Here, we assume $N_A \times J \geq I$ to ensure that all GUs can obtain communication services.

D. Communication Model

Let G_0 , G_1 , and G_2 denote the antenna gains of the ABS, GU, and MBS, respectively, β_0 denote the channel gain at a reference distance 1m. The channel gains from GU U_i to ABS A_j and from ABS A_j to MBS M_g at time slot t are given respectively by

$$h_{ij}(t) = \frac{G_0 G_1 \beta_0}{\|\mathbf{l}_i(t) - \mathbf{l}_j(t)\|_2^2} \quad (4)$$

$$h_{jg}(t) = \frac{G_0 G_2 \beta_0}{\|\mathbf{l}_j(t) - \mathbf{l}_g(t)\|_2^2}. \quad (5)$$

Let p denote the constant transmit power of all ABSs, B denote the channel bandwidth, and N_0 denote the received noise power. The transmission rate (bit/s) between ABS A_j and GU U_i is expressed as

$$r_{ij}(t) = x_{ij}(t) B \log_2 \left(1 + \frac{p \cdot h_{ij}(t)}{N_0} \right). \quad (6)$$

The transmission rate (bit/s) between ABS A_j and MBS M_g is expressed as

$$r_{jg}(t) = s_{jg}(t) B \log_2 \left(1 + \frac{p \cdot h_{jg}(t)}{N_0} \right). \quad (7)$$

E. Computation Model

ABS A_j has a computation task, $Y_j(t)$, and only a portion of the computation task, denoted as $Y_j^L(t)$, can be executed locally at ABS A_j due to the limited computing capability of the ABS. For delay improvement, the remaining portion of the task, denoted as $Y_j^O(t)$, is offloaded to MBS M_g that covers ABS A_j for computation. Full granularity in task partition is considered, where the task can be arbitrarily divided for local and remote executions [33], [34]. Let $C_j(t)$ denote the input data size (in bit) of computation task $Y_j(t)$, and $\gamma_j(t) \in [0, 1]$ denote the task offloading ratio of ABS A_j . Then, the task data sizes for local sub-task $Y_j^L(t)$ and offloaded sub-task $Y_j^O(t)$ are $C_j^L(t) = (1 - \gamma_j(t))C_j(t)$ and $C_j^O(t) = \gamma_j(t)C_j(t)$, respectively. ABS A_j has two queue buffers, where computation queue $Q_j^L(t)$ stores the local sub-tasks and transmission queue $Q_j^O(t)$ stores the offloaded sub-tasks. Let $f_j(t)$ denote the CPU-cycle frequency (in cycle/s) of ABS A_j and L_j denote the required CPU cycles per bit for computation, i.e., the processing density. Then, the local processing rate (in bit/s) of ABS A_j at time slot t is given by $D_j^L(t) = \frac{f_j(t)}{L_j}$. The queuing dynamics of $Q_j^L(t)$ and $Q_j^O(t)$ are modeled as

$$Q_j^L(t+1) = \max \{Q_j^L(t) + (1 - \gamma_j(t))C_j(t) - D_j^L(t)\delta_t, 0\} \quad (8)$$

$$Q_j^O(t+1) = \max \left\{ Q_j^O(t) + \gamma_j(t)C_j(t) - \sum_{g=1}^G r_{jg}(t)\delta_t, 0 \right\}. \quad (9)$$

Let $f_{jg}(t)$ denote the allocated CPU-cycle frequency at MBS M_g for processing offloaded sub-tasks from ABS A_j . The corresponding processing rate (in bit/s) of MBS M_g at time slot t is given by $D_j^S(t) = \frac{\sum_{g=1}^G s_{jg}(t)f_{jg}(t)}{L_j}$. MBS M_g maintains queue $Q_j^S(t)$ to store offloaded sub-tasks from ABS A_j with queuing state updated as

$$Q_j^S(t+1) = \max \{Q_j^S(t) + Z_j(t) - D_j^S(t)\delta_t, 0\} \quad (10)$$

where $Z_j(t) = \min \left(\sum_{g=1}^G r_{jg}(t)\delta_t, Q_j^O(t) + \gamma_j(t)C_j(t) \right)$ represents the task offloading data size from ABS A_j to MBS. Note that $Z_j(t)$ depends on transmission data size $\sum_{g=1}^G r_{jg}(t)\delta_t$, transmission queue length $Q_j^O(t)$, and offloaded sub-task data size $\gamma_j(t)C_j(t)$. When transmission data size $\sum_{g=1}^G r_{jg}(t)\delta_t$ is greater than the sum of transmission queue length $Q_j^O(t)$ and offloaded sub-task data size $\gamma_j(t)C_j(t)$, task offloading data size $Z_j(t)$ is determined by the sum of the queue length and offloaded sub-task data size $\sum_{g=1}^G r_{jg}(t)\delta_t$; otherwise, it depends on transmission data size $\sum_{g=1}^G r_{jg}(t)\delta_t$.

Specifically, if the sum of task offloading data size $Z_j(t)$ and queue length $Q_j^S(t)$ is less than corresponding processing data size $D_j^S(t)\delta_t$, the queue length at MBS M_g is 0.

Local sub-task $Y_j^L(t)$ experiences queuing delay for computation at ABS A_j . Offloaded sub-task $Y_j^O(t)$ experiences

queuing delay for both transmission from ABS to MBS and computation at the MBS. According to Little's law, the average computation queuing delay for local sub-task $Y_j^L(t)$ at ABS A_j is given by

$$T_j^L(t) = \frac{Q_j^L(t)}{D_j^L(t)}. \quad (11)$$

The average transmission queuing delay for offloaded sub-task $Y_j^O(t)$ from ABS A_j to MBS M_g is given by

$$T_j^O(t) = \frac{Q_j^O(t)}{\sum_{g=1}^G r_{jg}(t)}. \quad (12)$$

The average computation queuing delay for offloaded sub-task $Y_j^O(t)$ at MBS M_g is expressed as

$$T_j^S(t) = \frac{Q_j^S(t)}{D_j^S(t)}. \quad (13)$$

Note that the delay needed for returning the results back from MBS to ABS is ignored, due to the small data size [35].

F. Energy Consumption Model

The flight energy consumption of ABS A_j at time slot t is given by

$$E_j^{fl}(t) = P(v_j(t))\delta_t \quad (14)$$

where $P(v_j(t))$ is the propulsion power consumption at speed $v_j(t)$ and can be written as [36]

$$P(v_j(t)) = c_1 \|v_j(t)\|^3 + \frac{c_2}{\|v_j(t)\|} \left(1 + \frac{\|a_j(t)\|^2}{\alpha^2} \right) \quad (15)$$

where c_1 and c_2 are two parameters related to factors such as ABS's weight, wing area, and air density, and α is the acceleration of gravity [37]. Typically, the flight energy consumption by an ABS is much higher than that for computation and communication by two orders of magnitude [38]. By jointly considering the flight, computation, and communication energy consumption through weighting, some of existing studies consider energy consumption as the optimization objective [39], [40], [41], [42], while others use energy consumption as a part of the optimization objective [34], [43]. In our work, we focus on optimizing the average task queuing delay, while the total energy consumption is considered as a constraint. Hence, the energy consumption of computation and communication is not a focus here.

IV. PROBLEM FORMULATION

For the joint ABS computation offloading and ABS-assisted communication problem, the objective is to minimize the average task queuing delay while satisfying GUs' communication requirements subject to ABS-GU association constraints, computing constraints, and ABS mechanical constraints. Let $\mathcal{Z}_j(t)$ denote the set of GUs covered by ABS A_j at time slot t . The association between ABS A_j and GUs in $\mathcal{Z}_j(t)$ is represented by $\mathbf{x}_j(t) = \{x_{ij}(t), i \in \mathcal{Z}_j(t)\}$. Let $\mathbf{X} = \{\mathbf{x}_j(t), \forall j, t\}$,

$\mathbf{D} = \{\mathbf{v}_j(t), \mathbf{a}_j(t), \forall j, t\}$, and $\gamma = \{\gamma_j(t), \forall j, t\}$ denote the optimization variables corresponding to the ABS-GU association, ABS trajectory, and computation task offloading ratio, respectively. The optimization problem is formulated as

$$(P1) : \min_{\mathbf{X}, \mathbf{D}, \gamma} \quad \frac{1}{T} \sum_{t=1}^T \sum_{j=1}^J (T_j^L(t) + T_j^O(t) + T_j^S(t)) \quad (16a)$$

$$\text{s.t.} \quad E_j^{fl}(t) \leq E_{\max}, \forall j, t \quad (16b)$$

$$R_{\min} \leq \sum_{j=1}^J r_{ij}(t), \forall i, t \quad (16c)$$

$$\lim_{T \rightarrow \infty} \frac{\mathbb{E}[Q_j^L(t)]}{T} = 0, \forall j \quad (16d)$$

$$\lim_{T \rightarrow \infty} \frac{\mathbb{E}[Q_j^O(t)]}{T} = 0, \forall j \quad (16e)$$

$$\lim_{T \rightarrow \infty} \frac{\mathbb{E}[Q_j^S(t)]}{T} = 0, \forall j \quad (16f)$$

(1a) – (1f), (2), (3)

where $R_{\min}$ denotes the minimum transmission rate requirement for each GU and $E_{\max}$ denotes the per-slot flight energy budget of ABS A_j for preventing instances of excessive energy consumption at certain times and shortages at others [44], [45]. Constraints (16d)-(16f) ensure the task queue stability. The constraints can be categorized into three types: 1) GU communication constraints including (2), (3), and (16c), 2) computing constraints including (16d)-(16f), and 3) ABS mechanical constraints including (1a)-(1f) and (16b). The optimization problem is a MINLP. In addition, due to the propulsion energy consumption for the ABS, functions (1c), (1e), (16b), and (16c) are non-convex. Therefore, solving optimization problem (P1) is challenging. It is time inefficient and even intractable to find the global optimizer of such a non-convex problem. In the following section, we propose an approach to efficiently find a local optimum.

V. PROBLEM TRANSFORMATION AND SOLUTION

In this section, a joint DRL and Lyapunov optimization algorithm is presented to find a solution of problem (P1). Firstly, we transform problem (P1) into per-time-slot deterministic problems by using the Lyapunov-based method. Then, we adopt an actor-critic structure to solve the deterministic problem in each time slot. The actor module is a DNN that learns the optimal ABS-GU association action based on ABS-GU channel gains and queue backlogs of all ABSs. The critic module evaluates the ABS-GU association action by analytically solving a joint task offloading and trajectory planning problem.

A. Problem Transformation

The optimization problem (P1) is a long-term stochastic MINLP problem, which requires prohibitively high computational complexity. The DRL-based method is computationally feasible and efficient in solving such problems. However, the DRL-based methods fail to address the long-term performance requirements. Therefore, we apply Lyapunov optimization to

decouple the optimization problem (P1) into a series of short-term deterministic MINLP problems, which can be solved by DRL. Solely relying on the model-free DRL solution can lead to unstable performance and suffers from slow convergence or even divergence [13], [20], [46]. One approach is to have the model-free DRL for the optimization of partial variables (e.g., for optimizing binary variables), while using model-based optimization for the rest. Indeed, the integration of model-based optimization and model-free DRL can improve the robustness and convergence of the DRL framework [13], [47]. To solve a series of short-term deterministic MINLP problems, we propose a solution that integrates model-based optimization and model-free DRL based on the actor-critic structure.

Let $\mathbf{Q}(t) = (Q_j^L(t), Q_j^O(t), Q_j^S(t), \forall j)$ denote the combined queue vector of all ABSs. Define a Lyapunov function as

$$L(\mathbf{Q}(t)) \triangleq \frac{1}{2} \sum_{j=1}^J [Q_j^L(t)^2 + Q_j^O(t)^2 + Q_j^S(t)^2]. \quad (17)$$

Accordingly, a Lyapunov drift, $\Delta L(\mathbf{Q}(t))$, is given by

$$\Delta L(\mathbf{Q}(t)) \triangleq \mathbb{E}[L(\mathbf{Q}(t+1)) - L(\mathbf{Q}(t)) | \mathbf{Q}(t)]. \quad (18)$$

The Lyapunov drift-plus-penalty function is expressed as

$$\Delta L(\mathbf{Q}(t)) + V \mathbb{E} \left[\sum_{j=1}^J (T_j^L(t) + T_j^O(t) + T_j^S(t)) | \mathbf{Q}(t) \right] \quad (19)$$

where $V \geq 0$ is a weighting factor to balance optimization objective and queue stability.

Let $\mathbf{x}(t) = \{\mathbf{x}_1(t), \dots, \mathbf{x}_J(t), \dots, \mathbf{x}_J(t)\}$ denote the ABS-GU association decision, $\mathbf{v}(t) = \{\mathbf{v}_1(t), \dots, \mathbf{v}_J(t), \dots, \mathbf{v}_J(t)\}$ denote the ABS flight velocity decision, $\mathbf{a}(t) = \{\mathbf{a}_1(t), \dots, \mathbf{a}_J(t), \dots, \mathbf{a}_J(t)\}$ denote the ABS flight acceleration decision, and $\gamma(t) = \{\gamma_1(t), \dots, \gamma_J(t), \dots, \gamma_J(t)\}$ denote the ABS task offloading ratio decision, at time slot t . The stochastic problem, (P1), can be converted into a series of deterministic problems for each time slot, given by

$$(P2) : \min_{\mathbf{x}(t), \mathbf{v}(t), \mathbf{a}(t), \gamma(t)} \quad \Delta L(\mathbf{Q}(t)) + V \mathbb{E} \left[\sum_{j=1}^J (T_j^L(t) + T_j^O(t) + T_j^S(t)) | \mathbf{Q}(t) \right] \quad (19)$$

s.t. (1a) – (1f), (2), (3), (16b), (16c). (20)

Minimizing Lyapunov drift-plus-penalty function (19) needs the information of future time slots due to $\Delta L(\mathbf{Q}(t))$. To avoid involving future information, we derive and minimize its upper bound. Invoking a fundamental inequality, $\max\{x, 0\}^2 \leq x^2$, the upper bound of (19) is given by

$$\begin{aligned} & \Delta L(\mathbf{Q}(t)) + V \mathbb{E} \{T_j^L(t) + T_j^O(t) + T_j^S(t) | \mathbf{Q}(t)\} \\ & \leq C + V \mathbb{E} \{T_j^L(t) + T_j^O(t) + T_j^S(t) | \mathbf{Q}(t)\} \\ & \quad + \mathbb{E} \left\{ \sum_{j=1}^J \frac{1}{2} C_j^L(t)^2 + \frac{1}{2} C_j^O(t)^2 | \mathbf{Q}(t) \right\} \end{aligned}$$

$$\begin{aligned}
& + \mathbb{E} \left\{ \sum_{j=1}^J Q_j^L(t) [C_j^L(t) - D_j^L(t)\delta_t] | \mathbf{Q}(t) \right\} \\
& + \mathbb{E} \left\{ \sum_{j=1}^J Q_j^O(t) \left[C_j^O(t) - \sum_{g=1}^G r_{jg}(t)\delta_t \right] | \mathbf{Q}(t) \right\} \\
& + \mathbb{E} \left\{ \sum_{j=1}^J [Q_j^S(t) (Z_j(t) - D_j^S(t)\delta_t) \right. \\
& \quad \left. + \frac{1}{2} Z_j(t)^2] | \mathbf{Q}(t) \right\}
\end{aligned} \quad (21)$$

where C is a positive constant expressed as

$$C = \frac{1}{2} \sum_{j=1}^J \left[(D_j^L(t)\delta_t)^2 + \left(\sum_{g=1}^G r_{jg}(t)\delta_t \right)^2 + (D_j^S(t)\delta_t)^2 \right]. \quad (22)$$

A solution to problem (P2) can be obtained by minimizing the upper bound on the right hand side of (21) in each time slot t .

B. Problem Solutions

To minimize the upper bound in time slot t , we should find the optimal values of ABS-GU association decision $\mathbf{x}(t)$, continuous trajectory planning decision $\{\mathbf{v}(t), \mathbf{a}(t)\}$, and continuous task offloading ratio decision $\gamma(t)$, based on current observation $\mathbf{s}(t) = \{\mathbf{H}(t), \mathbf{Q}(t)\}$, consisting of combined channel gains $\mathbf{H}(t) = \{h_{ij}(t), h_{jg}(t), \forall i, j, g\}$ and combined queue vector $\mathbf{Q}(t)$. If ABS-GU association decision $\mathbf{x}(t)$ is given, problem (P2) is reduced to a joint task offloading ratio and trajectory planning optimization problem with decisions variables, $\{\mathbf{v}(t), \mathbf{a}(t), \gamma(t)\}$, which is a continuous non-convex optimization problem. Let $R(\mathbf{x}(t), \mathbf{s}(t))$ denote the optimal objective value of (P2) given offloading decision $\mathbf{x}(t)$ and observation $\mathbf{s}(t)$. We can find the optimal offloading decision $\mathbf{x}(t)^*$ as

$$(P3) : \mathbf{x}(t)^* = \arg \max [-R(\mathbf{x}(t), \mathbf{s}(t))]. \quad (23)$$

In general, obtaining $\mathbf{x}(t)^*$ requires enumerating $2^{I \times J}$ ABS-GU association decisions, which leads to significantly high computational complexity. To efficiently obtain $\mathbf{x}(t)^*$, we leverage the DRL technique and construct a policy, $\pi(\mathbf{x}(t)^* | \mathbf{s}(t))$, that maps from observation $\mathbf{s}(t)$ to ABS-GU association decision $\mathbf{x}(t)^*$. To obtain $R(\mathbf{x}(t), \mathbf{s}(t))$ given $\mathbf{x}(t)$, the joint task offloading ratio and trajectory planning optimization problem is decoupled into a task offloading ratio subproblem and a trajectory planning subproblem.

1) *ABS-GU Association Optimization*: As illustrated in Fig. 2, the proposed algorithm is built upon an actor-critic framework, which consists of an actor module, a critic module, and a policy update module. The actor module accepts input state $\mathbf{s}(t)$ and generates a set of candidate ABS-GU association actions. The critic module evaluates the potential actions and selects the best ABS-GU association decision. The policy update module improves the policy of the actor module over time. Details of each modules are provided as follows.

Actor Module: The actor module consists of action generation and action quantization. Let $\theta^\mu(t)$ denote the parameters

of actor network $\mu_\theta(s)$ in the actor module at time slot t . In the actor operation process, state $\mathbf{s}(t)$ is used as the input of actor network to produce ABS association decision action $\mathbf{x}'(t) \in [0, 1]^{I \times J}$, which is expressed as

$$\mathbf{x}'(t) = \mu_\theta(\mathbf{s}(t) | \theta^\mu(t)). \quad (24)$$

Let $\mathcal{F}(t)$ denote the set of all feasible ABS-GU association actions. In the actor quantization process, we calculate the Euclidean distance between $\mathbf{x}'(t)$ and all ABS-GU association actions in $\mathcal{F}(t)$, and select actions in $\mathcal{F}(t)$ corresponding to the top K smallest Euclidean distance as feasible candidate ABS-GU association actions $\hat{\mathbf{x}}(t) = \{\mathbf{x}_1(t), \dots, \mathbf{x}_K(t)\}$. The quantization function is given by

$$\hat{\mathbf{x}}(t) = \underset{\mathbf{x}_k(t) \in \mathcal{F}(t)}{\operatorname{argmin}}^K |\mathbf{x}_k(t) - \mathbf{x}'(t)|. \quad (25)$$

Critic Module: The critic module evaluates the candidate ABS-GU association actions in $\hat{\mathbf{x}}(t)$ and select best ABS-GU association action $\mathbf{x}_k(t)$. Actions $\mathbf{x}_k(t)$ with a high reward $-R(\mathbf{x}_k(t), \mathbf{s}(t))$ may occasionally sit closest to $\mathbf{x}'(t)$ even most ABS-GU association actions in $\hat{\mathbf{x}}(t)$ have a high reward [20], [48]. Different from a conventional actor-critic structure that uses a model-free DNN in the critic module to evaluate the actions, a model-based optimization method is used here to evaluate the actions in $\hat{\mathbf{x}}(t)$ by analytically solving a task offloading ratio subproblem and a trajectory planning subproblem. This enables the critic module to have an accurate evaluation of the ABS-GU association actions. We refine the choice of action by selecting the highest-reward ABS-GU association action according to

$$\mathbf{x}(t) = \underset{\mathbf{x}_k(t) \in \hat{\mathbf{x}}(t)}{\operatorname{argmax}} [-R(\mathbf{x}_k(t), \mathbf{s}(t))] \quad (26)$$

where $R(\mathbf{x}_k(t), \mathbf{s}(t))$ is obtained by optimizing the task offloading and trajectory planning given $\mathbf{x}_k(t)$ in problem (P2). Notice that the calculation of $R(\mathbf{x}_k(t), \mathbf{s}(t))$ is performed K times to obtain best action $\mathbf{x}(t)$. We introduce the detailed algorithm to obtain $R(\mathbf{x}_k(t), \mathbf{s}(t))$ in Subsection IV.C.

Policy Update Module: We use $(\mathbf{s}(t), \mathbf{x}(t))$ as a labeled input-output sample for updating the policy of the DNN in the actor module. The replay memory $\mathcal{D}$ is established to store data samples. In practice, with an initially empty replay memory, we start training the DNN after collecting more than $\frac{|\mathcal{D}|}{2}$ data samples. We randomly select a batch of data samples $\{(\mathbf{s}(\tau), \mathbf{x}(\tau)), \tau \in Z(t)\}$, where $Z(t)$ denotes the indices set of the selected samples. The parameter of the DNN is updated by minimizing average cross-entropy loss function, which is expressed as

$$\begin{aligned}
LS(\theta^\mu(t)) = & -\frac{1}{|Z(t)|} \sum_{\tau \in Z(t)} \left[\mathbf{x}(\tau)^T \log(\mu_\theta(\mathbf{s}(\tau) | \theta^\mu(t))) \right. \\
& \left. + (1 - \mathbf{x}(\tau))^T \log(1 - \mu_\theta(\mathbf{s}(\tau) | \theta^\mu(t))) \right]
\end{aligned} \quad (27)$$

where $|Z(t)|$ denotes the size of the sample batch, $(\cdot)^T$ denotes the transpose operator, and the log function denotes the element-wise logarithm operation of a vector. When the

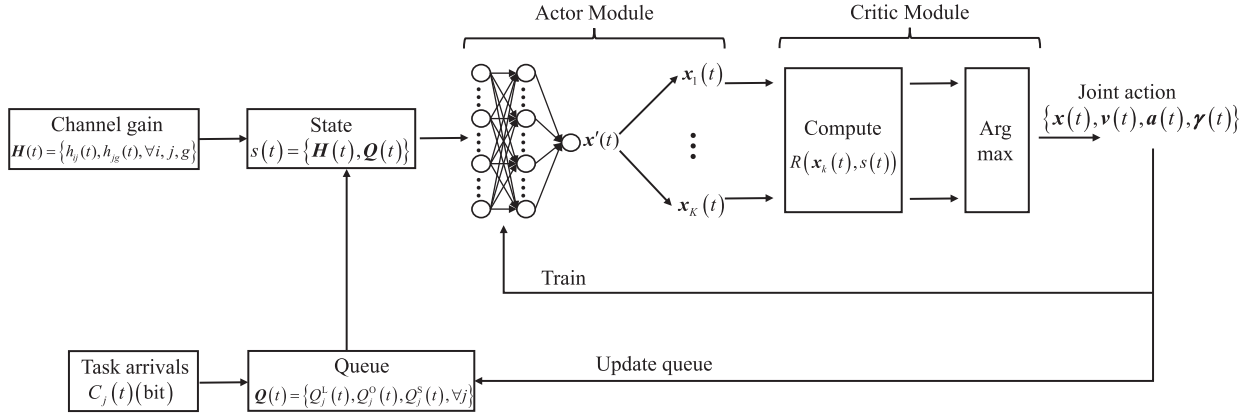


Fig. 2. The schematics of the proposed algorithm.

Algorithm 1 The Algorithm of JDL for Solving Problem**Require:** Initialize

input: Parameters $V, N_A, T, \delta_t, D_{\min}, v_{\max}, v_{\min}, a_{\max}, K$

output: Joint action $\{x(t), v(t), a(t), \gamma(t)\}$

Initialize actor network $\mu_\theta(s)$ with parameters $\theta^\mu(t)$ and empty replay memory $\mathcal{D}$;

```

1: while each episode do
2:   for  $t = 1, 2, \dots, T$  do
3:     Obtain observations  $s(t) = \{H(t), Q(t)\}$ ;
4:     Generate ABS association decision action  $x'(t) \in [0, 1]^{I \times J}$  using (24);
5:     Generate  $K$  feasible candidate ABS-GU association actions  $\hat{x}(t) = \{x_1(t), \dots, x_K(t)\}$  using (25);
6:     Compute  $R(x_k(t), s(t))$  by optimizing task offloading ratio decision  $\gamma(t)$  and trajectory planning decision  $\{v(t), a(t)\}$  in problem (P2) for each  $x_k(t)$ ;
7:     Select the highest-reward ABS-GU association action  $x(t)$  using (26) and execute the joint action  $\{x(t), v(t), a(t), \gamma(t)\}$ ;
8:     Update the replay memory by adding  $(s(t), x(t))$ ;
9:     if Replay memory has exceeded halfway capacity then
10:      Randomly select a batch of data samples from replay memory;
11:      Train actor network and update  $\theta^\mu(t)$ ;
12:   end if
13: end for
14: end while

```

training completes, we update the parameter of the actor module in the next time slot to $\theta^\mu(t+1)$. With the preceding actor-critic structure, the DNN consistently learns from the best and most recent state-action pairs, leading to a better policy that gradually approximates the optimal mapping to solve problem (P3). We summarize the pseudo-code of the algorithm for optimizing ABS-GU association in Algorithm 1.

2) *Task Offloading Ratio and Trajectory Planning Optimization:* Given the value of $x_k(t)$, we can decouple the joint task offloading and trajectory planning problem to compute $R(x_k(t), s(t))$.

Task Offloading Ratio Subproblem: Given values of $\{v(t), a(t), x_k(t)\}$, computation task offloading ratio $\gamma(t)$ can be obtained by minimizing the upper bound of (21), given by

(P2-1) :

$$\min_{\gamma(t)} \sum_{j=1}^J \left[\frac{1}{2} \gamma_j(t)^2 C_j(t)^2 + \frac{1}{2} (1 - \gamma_j(t))^2 C_j(t)^2 + Q_j^L(t) (1 - \gamma_j(t)) C_j(t) + Q_j^O(t) \gamma_j(t) C_j(t) - (1 - \gamma_j(t)) C_j(t) D_j^L(t) \delta_t - \sum_{g=1}^G \gamma_j(t) C_j(t) r_{jg}(t) \delta_t \right] \quad (28a)$$

$$\text{s.t. } 0 \leq \gamma_j(t) \leq 1, \forall j, t. \quad (28b)$$

From (28a), we observe that the computation task offloading ratios of all ABSs are independent. Hence, the task offloading ratios of different ABSs can be optimized separately. Accordingly, we obtain

$$\gamma_j(t) = \begin{cases} 0, & Q_j^O(t) \geq Q_j^L(t) + C_j(t) \\ 1, & Q_j^L(t) \geq Q_j^O(t) + C_j(t) \\ \frac{\phi_j}{2C_j(t)}, & \text{otherwise} \end{cases} \quad (29)$$

where $\phi_j = Q_j^L(t) + \sum_{g=1}^G r_{jg}(t) \delta_t - Q_j^O(t) - C_j(t) - D_j^L(t) \delta_t$.

In (29), $Q_j^O(t) \geq Q_j^L(t) + C_j(t)$ indicates that the offloaded executed tasks queue is shorter than queue length of local executed tasks. In this case, computation task $Y_j(t)$ is computed on the ABS. Inequality $Q_j^L(t) \geq Q_j^O(t) + C_j(t)$ indicates that computation task $Y_j(t)$ is computed on the MBS. When $Q_j^O(t) - C_j(t) < Q_j^L(t) < Q_j^O(t) + C_j(t)$, computation task $Y_j(t)$ is allocated to the MBS with task offloading ratio $\frac{\phi_j}{2C_j(t)}$.

Trajectory Planning Subproblem: With given values of $\{\gamma(t), \mathbf{x}_k(t)\}$, we optimize the ABS trajectories $\{\mathbf{v}_j(t), \mathbf{a}_j(t)\}$ and formulate the ABS trajectory planning subproblem as follows:

$$(P2-2) : \min_{\mathbf{v}(t), \mathbf{a}(t)} \sum_{j=1}^J \left[Q_j^S(t) Z_j(t) - \sum_{g=1}^G Q_j^O(t) r_{jg}(t) \delta_t + \frac{1}{2} Z_j(t)^2 \right] + \sum_{j=1}^J V [T_j^O(t) + T_j^S(t)]$$

s.t. (1a) – (1f), (16b), (16c). (30)

Problem (P2-2) is a non-convex problem due to the non-convex constraints (1c), (1e), (16b), and (16c). To construct a solvable approximation, non-negative auxiliary parameters $s_{jg}(t)$ and $T_j(t)$ are introduced. Problem (P2-2) can be transformed to

$$\min_{\mathbf{v}(t), \mathbf{a}(t), \{s_{jg}(t)\}} \sum_{j=1}^J \sum_{g=1}^G [(Q_j^S(t) - Q_j^O(t)) s_{jg}(t) \delta_t + \frac{1}{2} (s_{jg}(t) \delta_t)^2] + \sum_{j=1}^J V T_j(t) \quad (31a)$$

$$\text{s.t. } s_{jg}(t) \geq 0, \forall j, g \quad (31b)$$

$$s_{jg}(t) \leq r_{jg}(t), \forall j, g \quad (31c)$$

$$\frac{Q_j^O(t)}{s_{jg}(t)} + \frac{Q_j^S(t) \times L_j}{f_{jg}(t)} \leq T_j(t), \forall j, g \quad (31d)$$

$$Q_j^O(t) + \gamma_j(t) C_j(t) \geq s_{jg}(t) \delta_t, \forall j \quad (1a) - (1f), (16b), (16c), (31d) \quad (31e)$$

where (31c) and (16c) are concave [37]. Two constraints, (16b) and (1e), are tackled by using slack variable $\eta_j(t)$ that satisfy

$$E_j^{fl}(t) \leq \delta_t \left[c_1 \|\mathbf{v}_j(t)\|^3 + \frac{c_2}{\eta_j(t)} \left(1 + \frac{\|\mathbf{a}(t)\|^2}{\alpha^2} \right) \right] \quad (32)$$

$$\eta_j(t) \geq v_{\min}, \forall j, t \quad (33)$$

$$\|\mathbf{v}_j(t)\|^2 \geq \eta_j^2(t), \forall j, t. \quad (34)$$

It can be seen that (32) is convex, but (34) is a new non-convex constraint. To tackle this non-convex constraint, a local convex approximation is applied. Specifically, since $\|\mathbf{v}_j(t)\|^2$ is a convex and differentiable function with respect to $\mathbf{v}_j(t)$, for any local point $\mathbf{v}_j^{(r)}(t)$ obtained at the r -th iteration, we have

$$\|\mathbf{v}_j^{(r)}(t)\|^2 + 2(\mathbf{v}_j^{(r)}(t))^T (\mathbf{v}_j(t) - \mathbf{v}_j^{(r)}(t)) \geq \eta_j^2(t), \forall j, t. \quad (35)$$

It is obtained by the first-order Taylor expansion at the given points $\mathbf{v}_j^{(r)}(t)$ in the r -th iteration.

Considering the convexity of $\|\mathbf{l}_j(t) - \mathbf{l}_w(t)\|_2^2$, with ABS A_j location $\mathbf{l}_j^{(r)}(t)$ and ABS A_w location $\mathbf{l}_w^{(r)}(t)$ at the r -th iteration, we apply the SCA to reformulate (1c) to

$$D_{\min}^2 \leq \left(\|\mathbf{l}_j^{(r)}(t) - \mathbf{l}_w^{(r)}(t)\|_2^2 + 2 \|\mathbf{l}_j^{(r)}(t) - \mathbf{l}_w^{(r)}(t)\|_2 \right) \times \left(\mathbf{l}_j(t) - \mathbf{l}_j^{(r)}(t) + \mathbf{l}_w(t) - \mathbf{l}_w^{(r)}(t) \right), \forall j \neq w. \quad (36)$$

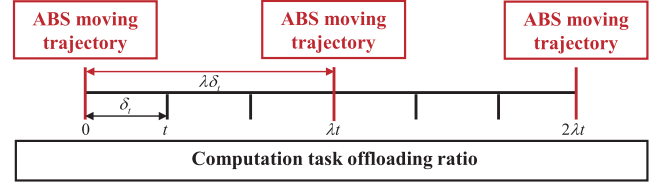


Fig. 3. Timeline of the two-timescale mechanism.

By applying the SCA method at the given points $\|\mathbf{l}_j^{(r)}(t) - \mathbf{l}_g(t)\|$, constraint (31c) is converted to

$$s_{jg}(t) \leq B \log_2 \left(1 + \frac{P G_0 G_2 \beta_0}{N_0 \|\mathbf{l}_j^{(r)}(t) - \mathbf{l}_g(t)\|_2^2} \right) - \frac{\delta(t) B P G_0 G_2 \beta_0}{N_0 \ln 2 \|\mathbf{l}_j^{(r)}(t) - \mathbf{l}_g(t)\|_2^2} \times \frac{1}{\|\mathbf{l}_j^{(r)}(t) - \mathbf{l}_g(t)\|_2^2 + \frac{P G_0 G_2 \beta_0}{N_0}} \times \left(\|\mathbf{l}_j^{(r)}(t) - \mathbf{l}_g(t)\|_2^2 + \|\mathbf{l}_j(t) - \mathbf{l}_g(t)\|_2^2 \right). \quad (37)$$

Based on the analysis above, problem (P2-2) is approximately transformed to

$$(P2-2-1) \min_{\mathbf{v}(t), \mathbf{a}(t), \{s_{jg}(t)\}} \sum_{j=1}^J \sum_{g=1}^G [(Q_j^S(t) - Q_j^O(t)) s_{jg}(t) \delta_t + \frac{1}{2} (s_{jg}(t) \delta_t)^2] + \sum_{j=1}^J V T_j(t)$$

s.t. (1a), (1b), (1d), (1f), (16c), (31b), (31d), (31e), (32), (33), (35), (36), (37). (38)

Problem (P2-2) is a non-convex problem due to the non-convex constraints (1c), (1e), (16b), and (16c). Constraint (1c) can be reformulated to (36), which is convex since it is linear with respect to $\mathbf{l}_j(t) - \mathbf{l}_w(t)$. By introducing slack variable $\eta_j(t)$, constraints (1e) and (16b) can be rewritten as (32), (33), and (35), which are convex [49]. By using non-negative auxiliary parameters $s_{jg}(t)$, (16c) can be rewritten as constraint (37), which is jointly convex of $\|\mathbf{l}_j^{(r)}(t) - \mathbf{l}_g(t)\|$. Therefore, (P2-2-1) is a convex optimization problem, which can be solved effectively with standard convex optimization solvers such as CVX. Denote $t_m = \lambda t$, where $\lambda \geq 1$ is a constant. The ABS moving trajectory is determined at time slot t_m , and the computation task offloading ratio is updated at time slot t , as shown in Fig. 3. Problem (P2-2-1) is a fractional minimization problem with a convex objective and convex constraints, which can be efficiently solved via the bisection method for fractional programming. Therefore, an efficient solution for problem (P2-2) can be achieved by successively updating the stationary point of problem (P2-2-1) at each iteration. We summarize the pseudo-code

Algorithm 2 The Algorithm for Optimizing Task Offloading Ratio and Trajectory Planning**Require: Initialize**

- input:** ABS-GU association decision $\mathbf{x}_k(t)$, $\mathbf{s}(t)$, and iteration tolerance $\xi > 0$;
 Initialize $\mathbf{l}^{(r)}(t)$, $\mathbf{v}^{(r)}(t)$, $R^{(r)}(\mathbf{x}_k(t), \mathbf{s}(t))$, and iteration index $r = 1$;
 1: **repeat**
 2: Generate $\gamma^*(t)$ using (29);
 3: Solve Problem (P2-2-1) with given $\{\gamma^*(t), \mathbf{x}_k(t)\}$ and obtain $\{\mathbf{v}^*(t), \mathbf{l}^*(t)\}$, $R^*(\mathbf{x}_k(t), \mathbf{s}(t))$;
 4: Update $\mathbf{l}^{(r+1)}(t) = \mathbf{l}^*(t)$, $\mathbf{v}^{(r+1)}(t) = \mathbf{v}^*(t)$ and $R^{(r+1)}(\mathbf{x}_k(t), \mathbf{s}(t)) = R^*(\mathbf{x}_k(t), \mathbf{s}(t))$;
 5: Update the iterative number $r = r + 1$;
 6: **until** the stopping criterion $|R^{(r+1)}(\mathbf{x}_k(t), \mathbf{s}(t)) - R^{(r)}(\mathbf{x}_k(t), \mathbf{s}(t))| \leq \xi$ is met;
 7: Obtain $\mathbf{v}^*(t)$, $\mathbf{a}^*(t)$, $\gamma^*(t)$, and $R^*(\mathbf{x}_k(t), \mathbf{s}(t))$.

Algorithm 3 Execution Stage of JDL**Require: Initialize**

- input:** The weights of actor network and execution time T ;
Output: Joint action $\{\mathbf{x}(t), \mathbf{v}(t), \mathbf{a}(t), \gamma(t)\}$;
 Initialize the task queue and locations of ABSs, MBSs, and GUs;
 1: **for** $t = 1, 2, \dots, T$ **do**
 2: Obtain observations $\mathbf{s}(t) = \{\mathbf{H}(t), \mathbf{Q}(t)\}$;
 3: Execute ABS association decision action $\mathbf{x}'(t) \in [0, 1]^{I \times J}$ based on the weights of actor network;
 4: Generate K feasible candidate ABS-GU association actions $\hat{\mathbf{x}}(t) = \{\mathbf{x}_1(t), \dots, \mathbf{x}_K(t)\}$ using (25);
 5: Compute $R(\mathbf{x}_k(t), \mathbf{s}(t))$ by optimizing task offloading ratio decision $\gamma(t)$ and trajectory planning decision $\{\mathbf{v}(t), \mathbf{a}(t)\}$ in problem (P2) for each $\mathbf{x}_k(t)$;
 6: Select the highest-reward ABS-GU association action $\mathbf{x}(t)$ using (26) and execute the joint action $\{\mathbf{x}(t), \mathbf{v}(t), \mathbf{a}(t), \gamma(t)\}$;
 7: **end for**

of the algorithm to solve task offloading ratio and trajectory planning problem in Algorithm 2. We consider a centralized-training-centralized-executing scheme to generate joint action $\{\mathbf{x}(t), \mathbf{v}(t), \mathbf{a}(t), \gamma(t)\}$ of all ABSs. Therefore, we select an ABS as a central controller, while the subproblems are solved and the JDL is trained and executed. The training process is shown in Algorithm 1 and the execution process is shown in Algorithm 3.

C. Convergence and Complexity Analysis

Convergence Analysis: The proposed JDL algorithm contains inner and outer structures. The outer structure is a Lyapunov-based optimization framework that transforms the long-term stochastic MINLP into a series of short-term deterministic MINLP problems. The inner structure is an integration of model-based optimization and model-free DRL

based on the actor-critic structure that is used to solve each deterministic MINLP problem. Hence, the convergence and complexity of the JDL mainly depend on the inner structure. The MINLP problem is decoupled into the ABS-GU association subproblem, ABS trajectory planning subproblem, and task offloading subproblem. The ABS-GU association subproblem is optimized in the actor module, and the ABS trajectory planning subproblem and task offloading subproblem are optimized in the critic module. In JDL, the outputs of ABS-GU association subproblem serve as the inputs of task offloading subproblem, and the outputs of task offloading subproblem serve as the inputs of ABS trajectory planning subproblem. The objective value is guaranteed to be non-increasing in consecutive iterations. Hence, the proposed JDL converges [39], [50].

Complexity Analysis: The computation complexity of JDL mainly stems from the ABS-GU association optimization and ABS trajectory planning optimization. For ABS-GU association optimization, we use a fully-connected DNN to process the input state and generate a set of candidate ABS-GU association actions. The dimension of input layer is $3J + (I + G)J$ and the dimension of output layer is IJ . Let ρ denote the number of hidden layers of the DNN and Φ denote the number of neurons in each hidden layer. The complexity for optimizing ABS-GU association is $O((3J + IJ + GJ) \times IJ \times \rho\Phi)$. The computation complexity of solving ABS trajectory planning optimization problem (P2-2-1) mainly depends on the second-order cone (SOC) constraints in (1d), (1f), (32), (35), (36), and (37) [37]. Specifically, problem (P2-2-1) involves $4J + J^2$ SOC constraints of dimension 3, JG SOC constraints of size 4, and $J^2 + 3J$ optimization variables [16], [51]. Therefore, the complexity of solving problem (P2-2-1) is roughly expressed as $\sqrt{2(4J + J^2 + JG)}(J^2 + 3J)(9(4J + J^2) + 16JG + (J^2 + 3J)^2)$, i.e., $O(\sqrt{2}J^7)$ [52].

VI. EXPERIMENTAL RESULTS

In this section, we evaluate the performance of the proposed joint DRL and Lyapunov optimization (JDL) algorithm. We deploy $J = 10$ ABSs and $I = 120$ GUs in a square area of $3000\text{m} \times 3000\text{m}$. We consider that an MBS is damaged with a certain probability after encountering disasters at the beginning of each cycle. This probability is contingent upon the severity of the disaster, which is set as 0.5 in the simulations. In the post-disaster scene, the surviving ground users are gathered in several safe areas. Accordingly, the GU distribution is obtained through a Poisson cluster process at the beginning of each cycle [10]. Setting $G = 2$ MBSs help execute tasks offloaded by ABSs and the flight altitude of ABS is 200m. Note that, one ABS offloaded task to one MBS in each time slot, and CPU frequency $f_g(t)$ of MBS M_g is evenly allocated for computing offloaded sub-task of ABSs in area S_g . The remaining parameters are given in Table II.

We consider a fully connected DNN consisting of one input layer, three hidden layers, and one output layer, where the number of neurons in each layer are set to 400, 300, and 100, respectively. The learning rate is 0.001, the batch size is 128,

TABLE II
SIMULATION PARAMETERS

Parameters	Values
Entire task period N	50s
Time interval δ_t	0.5s
Constant λ	4
Bandwidth B	5MHz
Maximum speed of each ABS $v_{\max}$	20m/s
Minimum speed of each ABS $v_{\min}$	3m/s
Maximum acceleration of each ABS $a_{\min}$	5m/s ²
Minimum secure distance $D_{\min}$	10m
ABS maximum available energy $E_{\max}$	4J
Channel power gain β_0	-21dBm [53]
Directional antenna gain of ABS G_0	9dBi [53]
Directional antenna gain of GU G_1	9dBi [53]
Directional antenna gain of MBS G_2	9dBi [53]
Noise power N_0	-169dBm/Hz
CPU frequency of ABS $f_j(t)$	1GHz [39]
CPU frequency of MBS $f_g(t)$	20GHz [54]
Required CPU cycles per bit L_j	1000 cycles/bit [39]
Parameter related to ABS c_1	9.26×10^{-4} [37]
Parameter related to ABS c_2	2250 [37]
Acceleration of gravity α	9.8m/s ² [37]

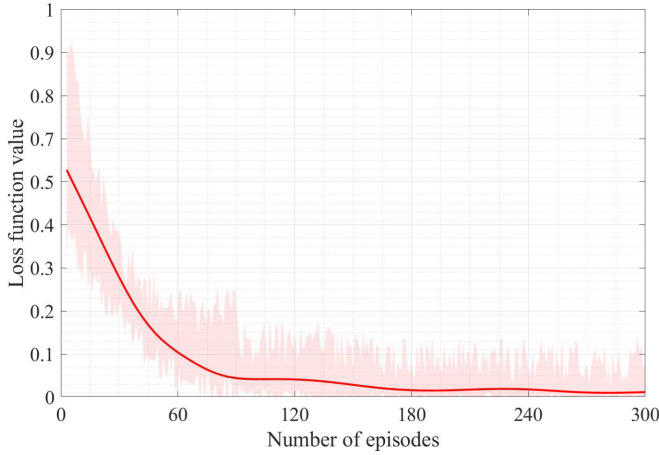


Fig. 4. The loss function value versus the number of episodes of the JDL algorithm.

the replay buffer size is 30000, the activation function is ReLu, and the Adam optimizer is used.

To show the effectiveness of the proposed algorithms, we consider the following benchmark algorithms for comparison.

SDQN: In [18], the authors propose a separated Q-network algorithm to jointly make optimal computation offloading decisions and trajectory planning for multi-UAV cooperative target search in an emergency network.

DTP-JDL: In [55], the authors propose a DRL-based trajectory planning method (DTP) to optimize the UAV trajectory in the UAV-assisted a post-disaster urban area with multiple mobile GUs for resilience scenario. We use the methods in JDL to optimize ABS-GU association and task offloading.

The loss function values during the training process are shown in Fig. 4. We have set multiple initialization values of DNN to obtain multiple loss function values during the training process. The red curve represents the average of loss function values, and the pink shadow indicates the range between the maximum and minimum loss function values.

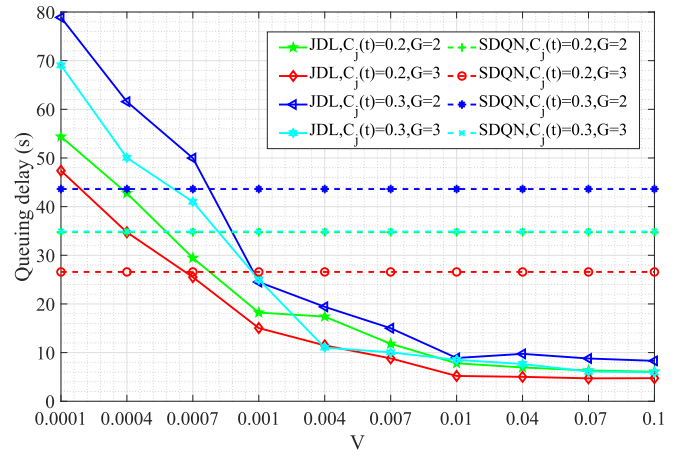


Fig. 5. Queuing delay of objective function versus parameter V .

It can be seen that the loss function value of JDL gradually converges as the number of episodes increases. Specifically, the achieved loss function value is less than 0.05 when the number of episodes exceeds 60.

The queuing delay is shown in Fig. 5. With the increase of V , the queuing delay decreases and converges, since the increasing value of V identifies the weight of queuing delay increasing in the objective function. Given parameter V , the queuing delay increases with the growth of computation task data size and the decrease in the number of MBSs. This phenomenon can be ascribed to the fact that a higher data size of computation task and a fewer number of MBSs result in an increase in the computation queuing delay of offloaded sub-tasks at the MBS, which leads to an increase in queuing delay. From Fig. 5, the queuing delay of JDL is smaller than SDQN since the critic module in JDL evaluates the ABS-GU association action by analytically solving the optimal task offloading and trajectory planning problem. Compared to the conventional actor-critic structure that uses a model-free DNN in the critic module, the proposed approach takes advantage of the model-based method to acquire an accurate evaluation of the action.

The comparisons of the queuing delay with different input data sizes $C_j(t)$ are shown in Fig. 6 at $J = 10$, $G = 2$, and $V = 0.01$. The JDL-Cir method means the trajectory planning of JDL using circular trajectory and the JDL-Ave means computation task offloading ratio $\gamma_j(t) = 0.5$, $\forall j, t$. With the increase of computation task data size, the queuing delay increases. The task offloading ratio decreases with the increase of input data size $C_j(t)$, leading to the growth of local computation task queue $Q_j^L(t)$, thereby increasing the queue delay.

The comparisons of the queuing delay with different numbers of ABSs are shown in Fig. 7 at $C_j(t) = 0.2\text{Mb}$, $G = 2$, and $V = 0.01$. It is evident that the queuing delay increases proportionally with the growth in the number of ABSs under the specified parameter. The increase in the number of ABSs leads to an increase in offloaded sub-task data size, resulting in an increase in the computation queuing delay at the MBSs. Consequently, the higher number of ABSs leads to a noticeable

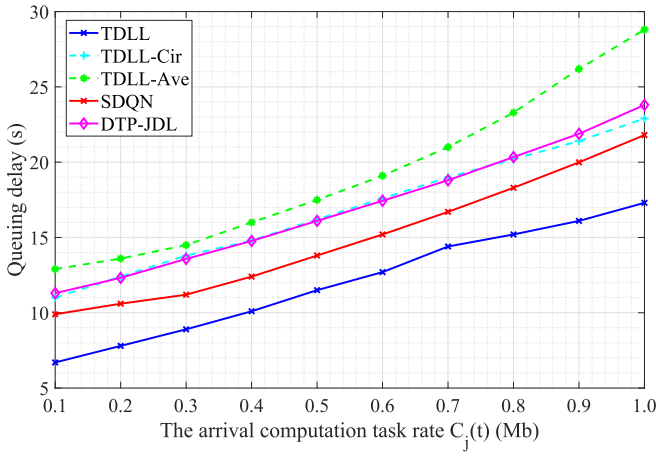
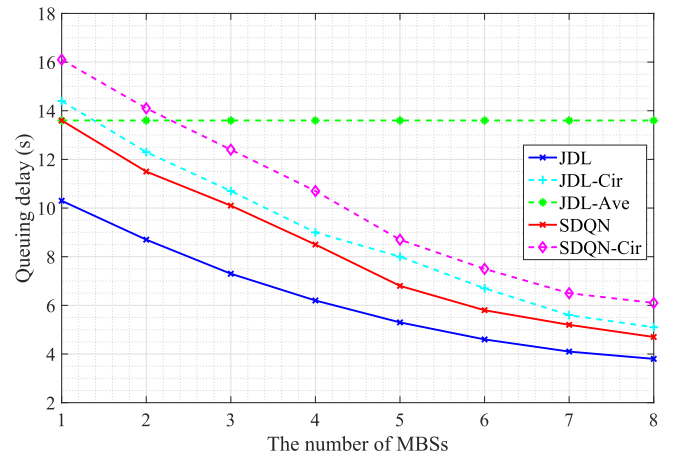

 Fig. 6. Queuing delay versus input data size of computation task $C_j(t)$.


Fig. 8. Queuing delay versus the numbers of MBSs.

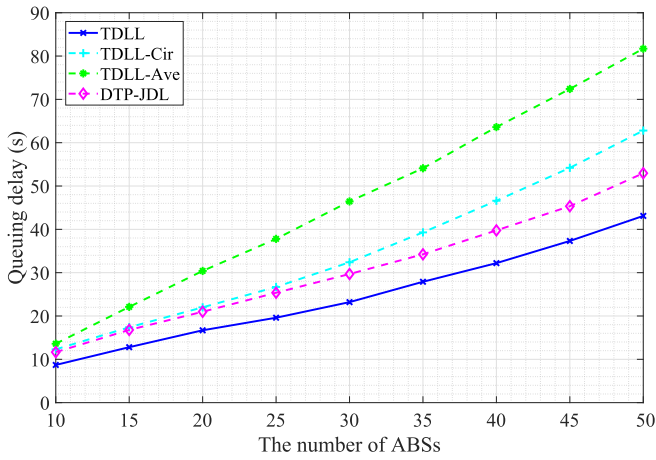
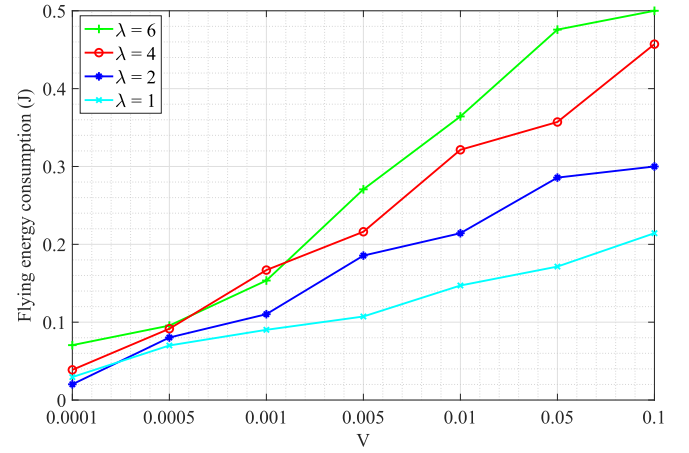


Fig. 7. Queuing delay versus the number of ABSs.


 Fig. 9. Flying energy consumption versus parameter V with different time scales.

increase in the queuing delay. Moreover, the DQN-based algorithms fail to obtain the ABS-GU association decision when $J = 20$, since the action space of DQN needs to include 2^{20} possible and available ABS association decisions to choose from. This is computationally infeasible.

The comparisons of the queuing delay with the different numbers of MBSs are shown in Fig. 8 at $C_j(t) = 0.2\text{Mb}$, $J = 10$, and $V = 0.01$. As the number of MBSs increases, there is a decrease in the queuing delay. With an increase in the number of MBSs, the task computing capability of all MBSs is enhanced, leading to a decrease in queuing delay $T_j^S(t)$. Meanwhile, the reduction in queuing delay of JDL gradually slows down as the number of MBSs increases. The reason is that the offloaded task from ABS to MBS is limited with limited maximum communication power.

Fig. 9 and Fig. 10 show the flying energy consumption and queuing delay, respectively, with different time-scales at $C_j(t) = 0.2\text{Mb}$, $J = 10$, and $G = 2$. From Fig. 9, the flying energy consumption is influenced by the time scale. When parameter V is smaller, the flying energy consumption increases with the increase in the time scale. Since the trajectory of ABS is not optimized in each time slot. Fig. 10 shows that the time scale has an impact on the queuing delay. As the

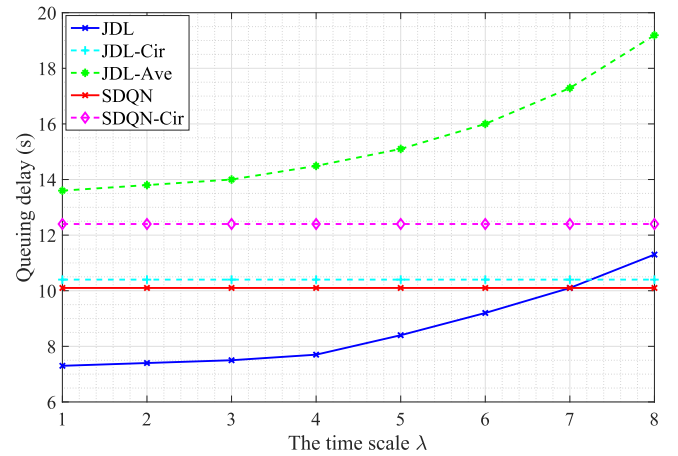


Fig. 10. Queuing delay versus time scales.

time scale increases, the queuing delay also increases. With a larger time scale, the intervals for trajectory optimization become larger, leading to a greater deviation between the ABS's trajectory and the optimal trajectory. Consequently, this results in an increase in the queuing delay.

Fig. 11 shows the average queuing length at ABSs and MBSs in a cycle at $C_j(t) = 0.5\text{Mb}$, $J = 10$, and $G = 2$. The

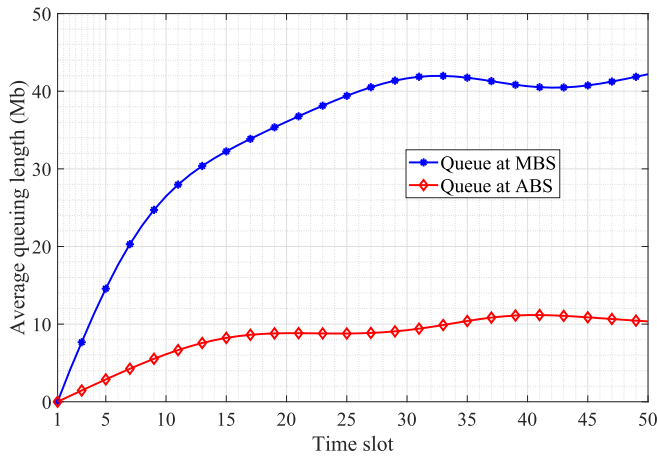


Fig. 11. Average queuing length at ABSs and MBSs.

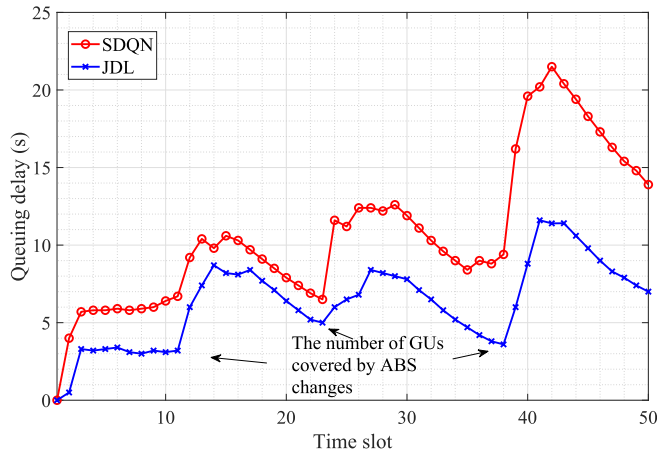


Fig. 12. Queuing delay between SDQN and JDL.

average queuing length at both ABSs and MBSs increases during the initialization phase and then gradually stabilizes, indicating the queue stability achieved by the JDL solution. Since the corresponding processing rate of MBS is faster than that of ABS, the task offloading ratio of ABS is large. Therefore, the average queuing length at MBS is longer than the average queuing length at ABS.

Fig. 12 shows the queuing delay between SDQN and JDL in a cycle. The queuing delay initially increases during the initialization phase and then stabilizes within a certain range. When the number of GUs covered by ABS changes, the queuing delay increases initially as the ABSs need to adjust the association strategy and plan new trajectory paths. Then, the queuing delay gradually decreases as the ABS-GU association, task offloading, and trajectory planning decisions gradually adjust. Compared to SDQN, the JDL combines model-based and DRL-based approaches, which can obtain ABS-GU association decisions by mapping Actor module output $\mathbf{x}'(t)$ to generated action set $\mathcal{F}(t)$. Therefore, the JDL reduces the impact of the number of GUs associated with ABS on the model structure.

Fig. 13 shows the queuing delay at ABS and MBS with different methods at $C_j(t) = 0.2\text{Mb}$ and $G = 2$. We consider the circular trajectory scheme as the benchmark, where the

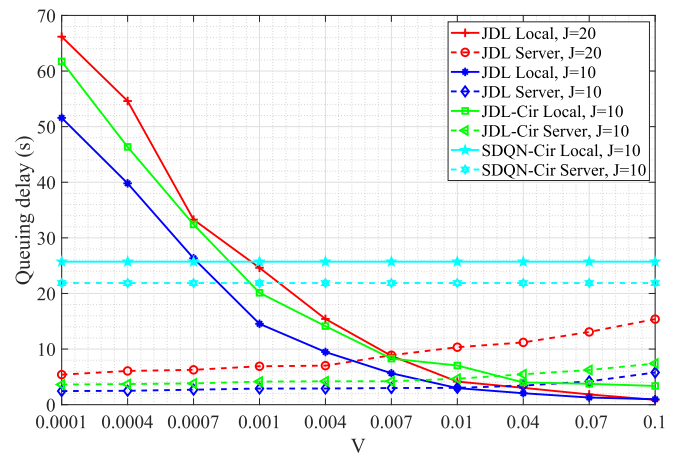
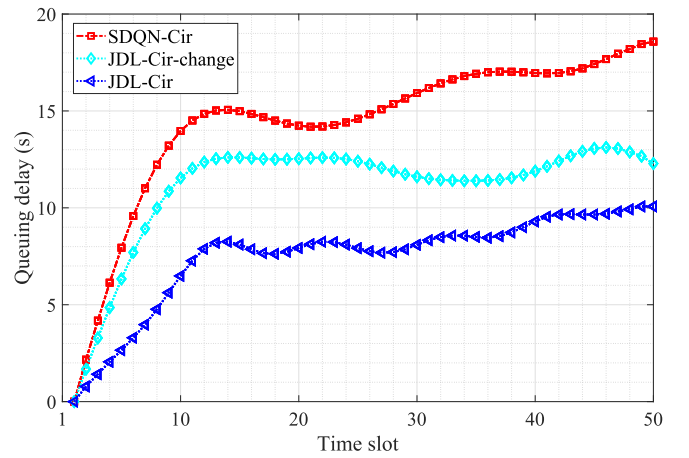
Fig. 13. Queuing delay at ABS and MBS versus parameter V .

Fig. 14. Queuing delay versus environment changes.

ABS moves around a circle within a cycle, with a radius of 2.5 km. The results in Fig. 13 show that the trajectory planning of JDL can reduce the queuing delay compared to the circular trajectory. Since the trajectory planning of ABS can shorten the distance between ABS and GU, and ABS and MBS, increase the task offloading transmission rate to reduce transmission. More portions of computation task data sizes $C_j^O(t)$ can be offloaded to MBS to reduce queuing delay.

To demonstrate JDL's capability on making real-time decisions based on fast model inference as the environment changes, we use the actual measured channel information in two scenarios [56]. The flight altitude and hover radius in the first and second scenarios are (2.5km, 2.0km), and (2.7km, 2.5km), respectively. Fig. 14 shows the queuing delay in a cycle at $C_j(t) = 0.3\text{Mb}$ and $G = 3$ as the environment changes. We train JDL-Cir-change model in the first scenario and then use it for execution in the second scenario. It can be seen that JDL-Cir-change effectively reduces the queuing delay compared to SDQN-Cir. The JDL-Cir is regarded as a benchmark, which is trained and executed in the second scenario.

Some real-time applications, such as real-time status monitoring based on channel state information (CSI), have a higher demand for accuracy in CSI. To show the application of JDL

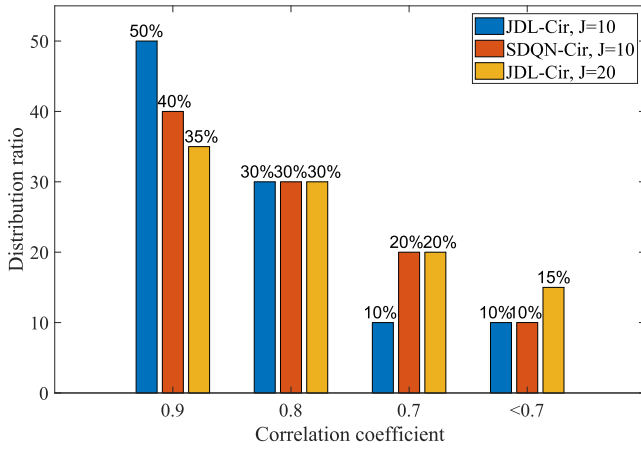


Fig. 15. Real-time application performance with different methods when the height is 2700 meters and the radius is 2500 meters.

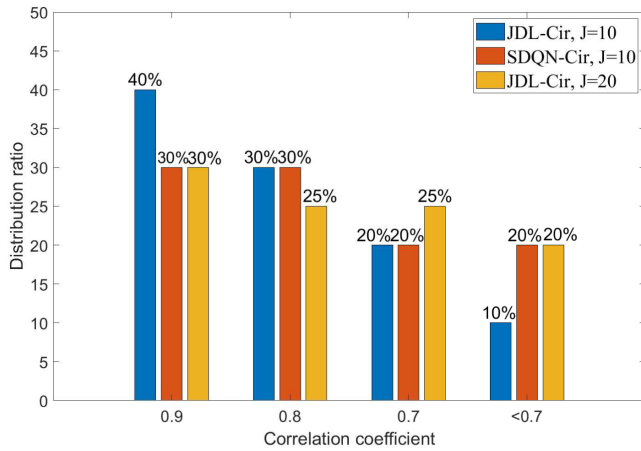


Fig. 16. Real-time application performance with different methods when the height is 2500 meters and the radius is 2000 meters.

can assist such delay-sensitive applications by minimizing queuing delay, we use the actual measurement data of correlation duration in [56]. The correlation duration is defined as the correlation coefficient between measured and outdated CSI is greater than a correlation coefficient threshold. Note that the queuing delay of ABS represents the delay when the CSI becomes outdated after the ABS completing task computation. If the queuing delay is shorter, the accuracy of the CSI for each ABS is higher. Conversely, if the queuing delay is longer, the accuracy of the CSI for each ABS is lower. In particular, CSI accuracy is manifested by the correlation coefficients of the channels. Compared to the queuing delay and correlation duration, we can obtain the CSI accuracy of each ABS. From Fig. 15, the JDL-Cir can minimize queuing delay with a high correlation coefficient compared with SDQN-Cir. Moreover, we test the performance of JDL-Cir in Fig. 16 under the parameter conditions where the height is 2500 meters and the radius is 2000 meters. Therefore, our proposed JDL can assist in real-time status monitoring based on CSI.

VII. CONCLUSION

In this paper, we have presented a model for a joint ABS-GU association, ABS trajectory planning, and task offloading

optimization problem in the ABS-assisted post-disaster rescue system, which is formulated as a stochastic MINLP. To minimize task computation queuing delay and ensure the GU communication rate, the MINLP minimization problem has been solved by integrating Lyapunov optimization and DRL based on actor-critic structure. Our research results provide valuable insights on joint ABS-GU association, ABS trajectory, and task offloading ratio design for computing rescue-related tasks and providing communication services to GUs. For future work, considering the uncertainty of user mobility and the time-variant ABS computation demand, we will investigate online resource management in an ABS-assisted post-disaster rescue system with wireless channel dynamics.

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